SIMATS ENGINEERING

ASSESSMENT TOOL 1-Pseudocode Writing Task

COURSE NAME: R PROGRAMMING **COURSE CODE: ITA04**

COURSE OUTCOME COVERED

CO1: Apply R programming fundamentals including data structures, vectors, matrices, arrays, and class types to perform mathematical operations and manage data effectively.
(Bloom Level -BL2- BL4)

Assessment Tool Weightage: 20%

Code of Conduct:

I 192321101 *(Name Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Prabhavathi R

ITA0406

R_Programming

CO1 - Assessment tool 1

Prabhavatti . R
192321101
BTech IT
14-07-2026

① variables and Data types Analysis

Pseudocode

```
> START
declare num as numeric = 25
declare name as character = "R Programming"
declare flage as logical = TRUE
sum = num + 15
product = num * 2
result = flag and FALSE
display "Numeric Value:", num
display "character value:", name
display "Logical value:", flag
display "sum :", sum
display "Product:", product
display "Logical Result :", result
> STOP
```

(2) Data structures Implementation Pseudocode

```
> start
create vector = [10, 20, 20, 40, 50]
Create List = ["R", 100, TRUE]
create Matrix = [[1,2], [3,4]]
display vector [2]
update vector [3] = 35
display List [1]
sum = sum(vector)
display "updated vector :", vector
display "Matrix :", Matrix
display "Vector Sum :", sum
display "Comparison:"
display "Vector_ same data type"
display "List_ Mixed datatype"
display "Matrix - Two dimensional data"
>STOP
```

(3) Matrix and Array Operations Pseudocode

>START

Create Matrix A = [[1,2], [3,4]]

Create Matrix B = [[5,6], [7,8]]

Matrix Add = Matrix X A + matrix X B

Matrix mul = Matrix X A × Matrix X B

Element wise = Matrix X A * matrix X B

Create Array 3D [2][2][2]

Display matrix X Add

Display Matrix X Mul

Display ElementWise

Display Array 3D

display "Matrix multiplication uses row - column rule".

display "Element - wise multiplicantion multiplies corresponding elements".

> STOP

(4) Data Frame Construction and Manipulation

Create Data Frame

Add column Id = [101, 102, 103]

Add Column Name = ["A", "B", "C"]

Add column Marks = [85, 90, 88]

Add column grade = ["A", "A+", "A"]

Delete Column grade

modify marks [2] = 95

display DataFrame

display Summary (DataFrame)

Stop

(5) Function and Logical operation. Pseudocode

> Start

function square (x) return x * x

end function

num = 6

answer = square (num)

Logical vector = [True, False, True, True]

Count = count (TRUE values in Logical vector)

display "Square:", answer

display "Number of True values:".

count

>stop

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

98/100